

ANALYSIS OF COMPLEX DYNAMICAL SYSTEMS

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1. INTRODUCTION AND MOTIVATION

Our project lies in the realm of *complex dynamical systems*. Roughly speaking, a dynamical system is a system that describes how a state evolves over time; the central questions are about the long-term behavior under repeated iterations of this rule.

Dynamics is the field that deals with change, with systems that evolve with time. Such a system is called a dynamical system. Formally, the definition of a dynamical system sufficient for the purposes of this project is presented in **Definition 1.0.1**. While there are several definitions of a dynamical system that are far more rigorous than what we have presented, our definition will suffice for the purposes of this project. We use the definition from [1].

Definition 1.0.1. A **dynamical system** is a pair (X, f) that consists of a set X (the *state space*) and a function $f : X \rightarrow X$ which describes how points of X change with repeated iterations of f .

There are several types of dynamical systems - linear dynamical systems, nonlinear dynamical systems, complex dynamical systems, and topological dynamical systems among several others. [2]

In this paper, we focus on *complex dynamical systems*. These are dynamical systems where the metric space X is a subset of the complex plane $\mathbb{C}$ and the mapping $f : X \rightarrow X$ is a complex-valued function. A simple example we use is iterating the quadratic polynomials

$$f_c(z) = z^2 + c, \quad z \in \mathbb{C}, c \in \mathbb{C}.$$

These functions look elementary, but their iterates exhibit remarkably rich behavior. When we repeatedly apply f_c to points in the complex plane, we see extremely beautiful geometric structures.

The main theme of this paper is to understand these functions and their iterations through a real-analysis lens, using tools such as limits, continuity, compactness, and derivatives. Along the way, we also discuss the connections between complex dynamics and fractals, and briefly touch on applications such as Newton's method and financial models.

2. BASICS OF DYNAMICAL SYSTEMS

In the introductory section, we introduced dynamical systems as pairs (X, f) with $f : X \rightarrow X$. In this section, we will describe certain properties and behaviors that points $x \in X$ exhibit with repeated iteration of f . However, in this section we will assume $X \subset \mathbb{C}$ so that we can measure distances using the standard metric $|\cdot|$, that is used in analysis.

Points exhibit certain behaviors under the iterative process, and the definitions below help describe these behaviors. We make use of these definitions to study specifics of Complex Dynamical Systems Later on. We first introduce a broader definition of one of these iterative process known as the Julia set and then introduce a few key definitions to motivate its construction/behavior.

Definition 2.0.1. A Julia Set is defined as the closure of the set of all repelling periodic points in a dynamical system [3].

This is one definition of a Julia set, and is equivalent to many other definitions. We introduce another definition of a Julia set in definition 3.0.4. which we will use as our main definition. This definition is used to motivate many other definitions in this section and adds another lens to intuit Julia sets from.

To define what a repelling periodic point is, we first need to define what the orbit of a point is.

Definition 2.0.2. The **orbit** of a point $x \in X$ is the sequence

$$\{x, f(x), f^{(2)}(x), f^{(3)}(x), \dots\}$$

where $f^{(n)}(x)$ represents the n^{th} iteration of f . We denote the orbit $O(x)$ [4]

From the definition we see that if we have two points x, x' and $x' \in O(x)$, then we have that $x' = f^n(x)$ so $\forall k \in \mathbb{N}, f^k(x') = f^{n+k}(x)$, so $O(x') \subseteq O(x)$. Now we will define what it means for an orbit to be periodic.

Definition 2.0.3. A **periodic point** $x \in X$ under a function f is a point where there exists $n \in \mathbb{N}$ such that $f^n(x) = x$. That is, after some number of iterations on x , the result is x , and the iterations are periodic [4].

A **fixed point** $x \in X$ under a function f is a point where $f(x) = x$ [4]. That is, it is a point that is invariant under the transformation f . A fixed point is a periodic point with period 1.

From this we can see that if x is periodic in the previous scenario, then $O(x) = O(x')$. Note that in dynamical systems, the set of all orbits partitions the entire space, and therefore distinct orbits never intersect. Finally we define what it means for a point to be repelling.

Definition 2.0.4. A **repelling point** is a point $x \in X$ under a function f such that there exists some $\delta > 0$ where for all points $t \in X$ with $|t - x| < \delta$, there exists some $N \in \mathbb{N}$ where for all $n \geq N$ we have that $|f^n(t) - x| > \delta$. That is, for all points in a neighborhood x , the trajectory of those points eventually leaves the neighborhood and they do not return.

As defined before all periodic repelling points are Julia set, but we also have that all non-periodic repelling points (and therefore all repelling points) are in the closure of the set of periodic repelling points. There are some other types of points we classify based on the behavior of orbits near them, most notably attractors.

Definition 2.0.5. An **attractor** is a subset A of the phase space of a dynamical system with the following properties: if $x \in A$ then $\forall n \in \mathbb{N}, f^n(x) \in A$. There exists a neighborhood of A called the **basin of attraction** and denoted $B(A)$ is the set of all x such that $\lim_{n \rightarrow \infty} f^n(x) \in A$. Finally, there is no proper subset of A having the first two properties [5].

Attractors can be thought of as the opposite of repellers, and basins of attraction make up the vast majority of space. For example, if the basin of infinity is an attractive basin, then it makes up all of space not enclosed within the Julia set. It is the complement of the filled Julia set, which will be defined shortly. There are also often one or many basins of attraction in the filled in Julia set.

3. COMPLEX DYNAMICAL SYSTEMS

Now we dive into the heart of our project where we focus on dynamical systems where $X = \mathbb{C}$ with the metric $d(z, w) = |z - w|$. To first go through and conduct any analysis, we need to first define a notion of differentiability for complex-valued functions or $f : \mathbb{C} \rightarrow \mathbb{C}$.

Definition 3.0.1. Let $U \subseteq \mathbb{C}$ be an open set and $f : U \rightarrow \mathbb{C}$ be a function. f is **complex differentiable** at $z_0 \in U$ if the limit

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

exists and is the same irrespective of the direction in which z approaches z_0 in the complex plane [6].

This is a ‘complex analysis’ equivalent to differentiability that we study in real analysis.

A function f is **holomorphic** (or analytic on U) if f is complex differentiable at every $z \in U$ [6].

This is defined in a very similar manner to the derivative from real analysis, but the condition is actually stronger since in $\mathbb{R}$ there are only two directions from which we can approach a point but on $\mathbb{C}$ there are many different ways in which we can approach a point and we need z to approach z_0 in all directions.

From complex analysis we can prove many powerful theorems about holomorphic functions: for example, they can be represented locally by convergent power series which we learned in MA 440. However, for this project we will only need the following basic facts:

- Polynomials with complex coefficients are holomorphic on all of $\mathbb{C}$. [4][1]
- The derivative of a complex polynomial is obtained in the same way as in one-variable calculus, by using the usual rules of differentiation. [6]

Particularly, for the quadratic polynomial $f_c(z) = z^2 + c$ where $c \in \mathbb{C}$ we have

$$f'_c(z) = 2z, \quad z \in \mathbb{C}$$

So to compute derivatives, it is very similar to how we do it for real-valued functions.

One key property of points in complex dynamical systems which we focus on is the *stability* of these points under iteration.

Stability of fixed points. Recall from the dynamical systems section that a *fixed point* of a map f is a point z_0 with $f(z_0) = z_0$. We now use fixed points to extend concepts in Complex Dynamical Systems. [7]

Definition 3.0.2. Let $U \subset \mathbb{C}$. We consider $f : U \rightarrow \mathbb{C}$ be a map and let $z_0 \in U$ be a fixed point.

- We say that z_0 is **stable** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $|z - z_0| < \delta$ we have

$$|f^{(n)}(z) - z_0| < \varepsilon \quad \text{for all } n \geq 0.$$

In other words, orbits that start close to z_0 stay close for all time.

- We say that z_0 is **asymptotically stable** (or **attracting**) if it is stable and, in addition,

$$f^{(n)}(z) \rightarrow z_0 \quad \text{as } n \rightarrow \infty$$

for all z in some neighborhood of z_0 .

- We say that z_0 is **repelling** if there is a neighborhood of z_0 such that every point $z \neq z_0$ in that neighborhood is eventually pushed away from z_0 under iteration.

A key connection between the theory and dynamical systems is that for holomorphic functions, the derivative at a fixed point entirely dictates the local behavior [2].

Theorem 3.0.3 (Stability via the derivative). *Let $f : U \rightarrow \mathbb{C}$ be holomorphic on an open set $U \subset \mathbb{C}$ and let $z_0 \in U$ be a fixed point, $f(z_0) = z_0$. Set $\lambda = f'(z_0)$. Then:*

- If $|\lambda| < 1$, then z_0 is asymptotically stable (attracting). [7]
- If $|\lambda| > 1$, then z_0 is repelling. [7]
- If $|\lambda| = 1$, the fixed point is **neutral**: the derivative test alone does not decide stability, and more subtle behavior can occur. [7]

Idea of proof. Near z_0 the holomorphic map f can be approximated by its linearization

$$f(z) \approx z_0 + f'(z_0)(z - z_0),$$

so the distance to z_0 is approximately multiplied by $|\lambda|$ at each step since we defined $\lambda = f'(z_0)$. Using basic tools from complex analysis (in particular, integrating f' along line segments), one can make this precise and show that:

- If $|\lambda| < 1$, then for z sufficiently close to z_0 there is a constant $L < 1$ such that

$$|f(z) - z_0| \leq L|z - z_0|.$$

This makes f a contraction on a small disk around z_0 , and repeated iteration forces $f^{(n)}(z) \rightarrow z_0$.

- If $|\lambda| > 1$, then the inverse map f^{-1} is defined and holomorphic on a small neighborhood of z_0 , and its derivative satisfies $|(f^{-1})'(z_0)| = 1/|\lambda| < 1$. By the previous case, z_0 is attracting for f^{-1} , which means it is repelling for f . [7]

The neutral case $|\lambda| = 1$ is more delicate and we do not treat it here. □

In this report, we present a rough idea of the proof, however the entire proof requires significant knowledge of complex analysis which can be found in *Milnor's Dynamics in One Complex Variable* book [7], in the chapter about **Local Fixed Point Theory**.

Quadratic maps and stability. We now return to the family of quadratic maps

$$f_c(z) = z^2 + c, \quad c \in \mathbb{C}.$$

A fixed point z_0 of f_c satisfies

$$z_0 = z_0^2 + c \iff z_0^2 - z_0 + c = 0,$$

so the fixed points are the roots of this quadratic:

$$z_0 = \frac{1 \pm \sqrt{1 - 4c}}{2}.$$

The derivative is

$$f'_c(z) = 2z,$$

so at a fixed point z_0 the multiplier is $\lambda = f'_c(z_0) = 2z_0$. By the Theorem about stability:

- If $|2z_0| < 1$, then z_0 is an attracting fixed point of f_c .
- If $|2z_0| > 1$, then z_0 is repelling.

As we vary the parameter c , the locations of the fixed points change, and so does their stability. This changing local behavior is closely linked to the geometry of what we can call a *Julia set*, which we will be discussing next.

For each fixed $c \in \mathbb{C}$, iterating f_c partitions the plane into points whose orbits remain bounded and points whose orbits escape to infinity.

Definition 3.0.4. Fix $c \in \mathbb{C}$ and $f_c(z) = z^2 + c$.

- The **filled Julia set** of f_c is

$$K(f_c) = \{z \in \mathbb{C} : \{f_c^n(z)\}_{n \geq 0} \text{ is bounded}\}.$$

[4]

- The **Julia set** of f_c is its boundary

$$J(f_c) = \partial K(f_c).$$

[4]

Intuitively, $K(f_c)$ consists of all points whose orbits do *not* escape to infinity under iteration of f_c , while $J(f_c)$ is the “edge” separating stable bounded behavior from escape to infinity. The Julia set $J(f_c)$ is where the dynamics are most sensitive to initial conditions, and it is this boundary which typically exhibits rich fractal structure.

In the next section, we introduce fractals more systematically (starting from the middle-third Cantor set) and then return to Julia sets to view them through this fractal lens.

4. FRACTALS AND JULIA SETS

Julia sets are a specific subset of mathematical objects known as fractals. At its core a fractal is a geometric shape that maintains structure at arbitrarily small scales, with the characteristic that its fractal dimension is strictly greater than the topological dimension. Intuitively, the fractal dimension refers to how many copies of an object are needed to scale it by a certain factor. This is explicitly shown in the formula $N = r^D$, where N is the number of total objects needed, r is the factor we want to scale by and D is the fractal dimension. In $\mathbb{R}^1$ if we want to scale an object by 2, we need $2^1 = 2$ copies of our object. In $\mathbb{R}^2$ if we want to scale our object (in this case a line) by 2, we need $2^2 = 4$ copies of our object.

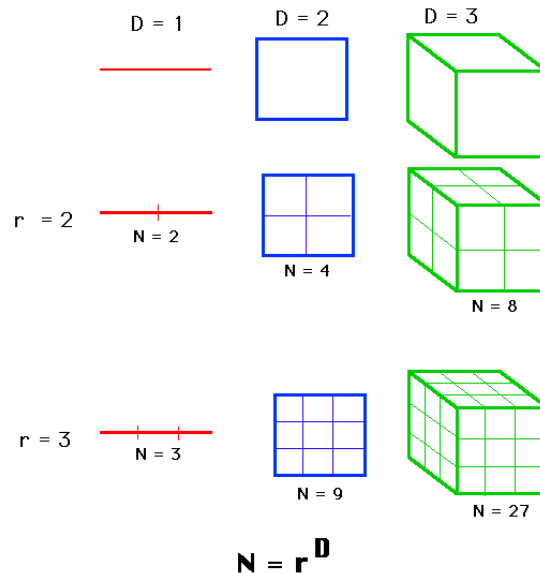


FIGURE 1. Examples of increasing fractal dimension

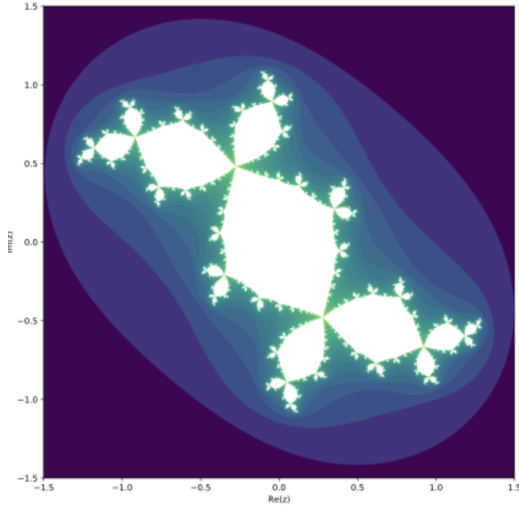


FIGURE 3. The Douady Rabbit ($c \approx -0.12256 + 0.74486i$) is a connected Julia set [9]

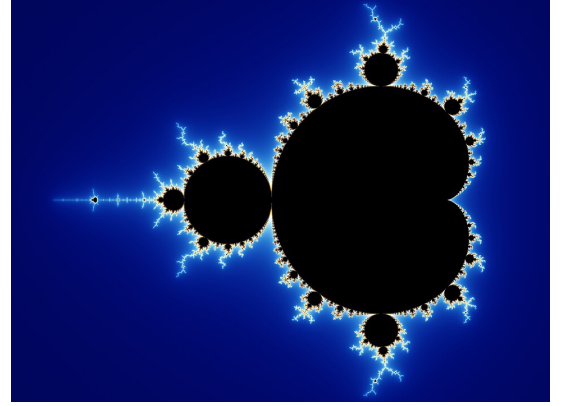


FIGURE 4. The Mandelbrot Set [8]

Determining if a Julia set is connected is surprisingly simple. A constant c is in the Mandelbrot set if and only if there exists some constant $K \in \mathbb{Z}$ such that for a function $f(z) = z^2 + c$ then $\forall n \in \mathbb{N}$ we have $f^n(0) \leq K$. Therefore we only need to analyze a single orbit for each quadratic Julia set to determine if it is connected.

5. APPLICATIONS OF COMPLEX DYNAMICS

5.1. Application: Newton's method of finding roots. A popular numerical algorithm which fits into complex dynamical systems is *Newton's method* to compute roots of a function. Let us consider a polynomial $P(z)$. Newton's method defines the *Newton Map* [10]

$$N_P(z) = z - \frac{P(z)}{P'(z)}$$

We start with an initial guess $z_0 \in \mathbb{C}$ and we carry out the following iterations

$$z_{n+1} = N_P(z_n), \quad n = 0, 1, 2, \dots$$

We can see that this is *Dynamical System* $(\mathbb{C}, N_P)$ [11].

Let x be the root of the polynomial P . Each x is a fixed point of N_P since $P(x) = 0 \implies N_P(x) = x$. For several polynomials, these fixed points are attracting, and as we defined earlier, the complex plane divides these into *basins of attractions*

$$\{z_0 \in \mathbb{C} : N_P^{(n)}(z_0) \rightarrow x \text{ as } n \rightarrow \infty\}$$

If we associate a color for each initial point z_0 depending on which root the Newton's method computation converges to, the complex plane will break into different colored regions. Each region will be a basin of attraction for a root which gives rise to a **fractal** structure since the boundary between these basins is the Julia set of the Newton Map.

To further understand this application, we ran a numerical experiment to compute and show the Newton Fractal of $P(z) = z^3 - 1$ which was done entirely using Python code.

We consider the polynomial $P(z) = z^3 - 1$ and we know it has three roots. Using this fact, we create a square region in the complex plane and apply Newton's method from each initial point. Then we color each point based on which root the *orbit* of the point converges to and how many iterations it takes.

The image below shows three distinct regions, separated by a fractal boundary that approximates the Julia Set of N_P .

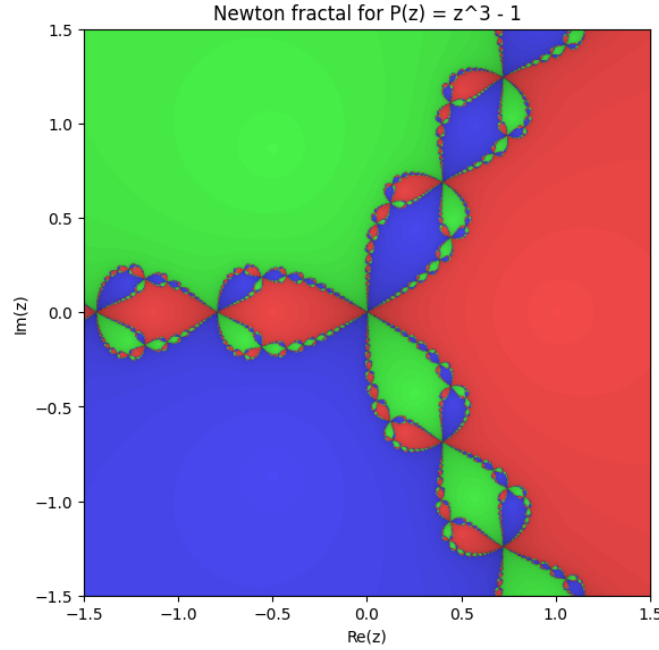


FIGURE 5. Newton fractal for $P(z) = z^3 - 1$

We implemented this experiment in Python; the complete code and an interactive notebook are available at the following link:

<https://colab.research.google.com/drive/12MeIg6TKC1YVyFn74raUFVA9pjf-g47M?usp=sharing>

5.2. Connection to Financial Markets. In financial markets, a common assumption in many popular models is the Efficient Market Hypothesis (EMH). EMH states that the traded value of any asset perfectly reflects its fair value and thus any potential for arbitrage is removed. Geometric Brownian Motion is used to model the "random walk" nature exhibited in stock prices under this framework. While not a perfect model, it keeps prices positive, volatility proportional to price and models returns as being normally distributed which are all characteristics seen within financial markets. However, the self-similarity characteristics that are present within Fractals is present in a similar capacity within financial markets and as such frameworks that can model that are desirable. Development in this area led to something called the "Fractal Market Hypothesis" [12]. At its core is the observation that the distribution of returns exhibits similar qualities across different time scales. Fractional Brownian Motion (fBm) models this behavior through a parameter known as the Hurst coefficient denoted H where $0 < H < 1$. Higher values of H exhibit smoother trends and lower volatility.

The covariance function $\mathbb{E}[B_H(s)B_H(t)] = \frac{1}{2} \left(t^{2H} + s^{2H} - |t-s|^{2H} \right)$ [13] defines this process and the Hurst coefficient is directly related to the Fractal dimension D when $1 < D < 2$ via the equality $D = 2 - H$.

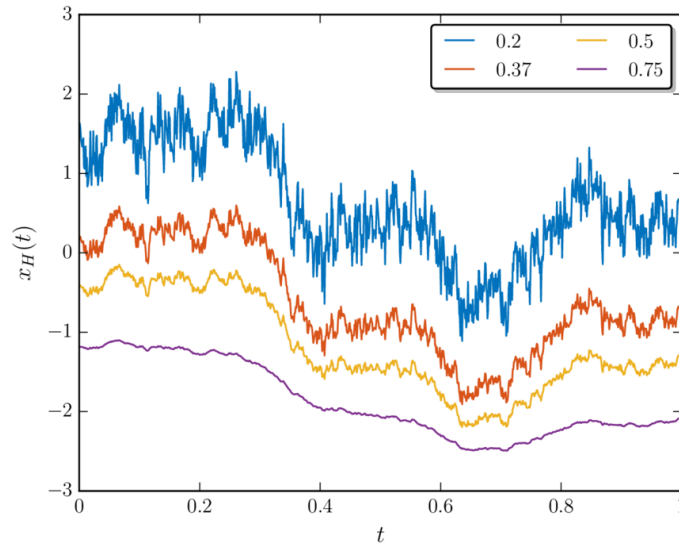


FIGURE 6. Fractional BM with different Hurst Parameters

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